SyllabusSherlock

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Abstract—**Students often struggle with textbook-based learning because the learning process is not always personalized to their needs. To solve this, we developed SyllabusSherlock, an AI-assisted learning system based on the NCTB curriculum for Bangladeshi students. It uses a curriculum-based Knowledge Graph to connect topics and their prerequisite relationships while retrieval and generative AI are used to give relevant learning content and questions. NCTB textbook content is organized into books, chapters, pages, and topics with prerequisite connections linking related concepts. We evaluate the generated questions using an LLM-based judge that checks their quality and alignment with the grounded textbook content. The evaluation shows promising results in relevance and textbook alignment while also highlighting areas where question quality can be improved. SyllabusSherlock shows how these techniques can be combined to provide more personalized and curriculum-based learning support for Bangladeshi students.**

Index Terms—**Knowledge Graph, Personalized Learning, Intelligent Tutoring System, Generative AI, NCTB Curriculum**

I. Introduction

Open with the motivation for your problem, cite prior foundational work [1], and close with a clear statement of your contributions:

- Contribution one.
- Contribution two.
- Contribution three.

A short paragraph describing the structure of the rest of the paper ("The remainder of this paper is organized as follows...") is optional but common in this style.

II. Related Work

A. Knowledge Graphs and Educational Knowledge Representation

Knowledge Graphs (KGs) provide a structured way to represent information and relationships between concepts [1], [2]. Previous studies have used KGs to improve the understanding of AI systems and causal reasoning [1], [2]. However, traditional Retrieval-Augmented Generation (RAG) can struggle with complex documents as similarity-based retrieval may not capture hierarchical dependencies or temporal relationships [1]. To address these limitations, one study proposed agentic KGs with recursive crawling to follow interconnected citations and reported up to 70 % higher accuracy than standard vector-based systems [1]. LLMs have also been integrated with KGs to provide additional information when structural data are sparses [1].However, domain-specific KGs for local educational curricula are still limited with much of the existing work focusing on transportation [1] and general knowledge sources [2]. In SyllabusSherlock, a topic-prerequisite Knowledge Graph is used to represent the structure and relationships in NCTB textbooks.

B. Intelligent Tutoring Systems and Personalized Learning

Intelligent Tutoring Systems (ITS) use Bayesian Knowledge Tracing (BKT) and performance modeling to personalize learning support [1], [3]. Research indicates that ITS can achieve learning gains comparable to one-on-one human tutoring [1], [3]. The Intelligent Multi-Tutoring System (IMTS) combines machine and human tutoring to improve learner retention [1]. Recent ITS research focuses on ”productive struggle” where students receive hints instead of direct solutions [4].At the same time, conventional Learning Management Systems (LMS) are mainly used for administrative tasks and often provide limited real-time instructional supports [4]. Our system uses BKT to track topic mastery and provide personalized conversational support for middle-school learners.

C. Adaptive Assessment and Automated Question Generation

Automated Question Generation (AQG) has increasingly incorporated RAG to generate questions from source documents and reduce unsupported AI responses [1], [5], [6]. Some studies also use ensemble learning to reduce repeated topics and increase the diversity of generated test sets [1]. Some systems adjust quiz difficulty based on performance trends but this is often done at the class level rather than for individual students [1], [5]. Evaluating higher-order thinking skills is another challenge for automated systems [1], [7]. Our project uses textbook-based question generation to adapt quiz difficulty to individual students' performance and mastery levels.

D. Learning Analytics, Retention and Gamification

Learning analytics use student data to identify learning patterns and support personalized interventions [8], [9]. Studies in STEM education have found a strong positive correlation (R^2=0.76) between interaction time with tutoring systems and mastery of fundamental conceptss [1]. Empirical evidence suggests that real-time feedback in adaptive systems can improve student post-test scores by up to 16.9 % [1].To improve retention, computational models can use performance history to schedule spaced practice based on individual forgetting dynamics. [4]. On the other hand, gamification research shows that integrating points and badges can enhance student engagement and academic self-efficacy [1]. However, many studies evaluate gamification over short periods making it's long-term effect on student motivation difficult to determine. [1], [10]. Our project addresses this by using performance data to drive spaced repetition scheduling while maintaining engagement through XP, streaks, and character avatars.

E. Generative AI in Education

Generative AI has been explored for tasks such as natural-language explanation and identifying student misconceptions [7]. Recent studies have shown that LLMs can identify and explain Bengali-specific grammatical errors. However, using these models with minors raises concerns about reliability and hallucinated explanations [8], [7]. This creates a need for systems that manage AI responses based on the learning context and safety requirements [7]. Our system uses a recent model within an RAG architecture to ground tutoring responses in NCTB textbook content. The retrieved textbook content helps keep responses aligned with the curriculum, while the Knowledge Graph provides the learning roadmap for students.

III. Data Collection and Description

SyllabusSherlock uses scanned NCTB textbooks for class 6-8 initially focusing on Mathematics and Science as it's primary source of educational content. These textbooks contain the content of the curriculum which are organized by topics, retrieved from the content, used for AI learning and generate assessments. Collected textbooks are organized according to their class, subject, book, chapter and page structure.

The PDFs of the textbook are scanned and processed through an OCR pipeline to obtain machine-readable Bengali text from the scanned pages. The text is then de-noised by OCR to remove OCR noise and segmented to smaller chunks for semantic retrieval. Content is stored as cleaned pages and content chunks and curriculum structure is kept separately in the Knowledge Graph. The resulting chunks of text are then embedded as vectors and indexed in FAISS for retrieval. This enables the system to generate semantic similarity based relevant content from the textbook. In parallel, the Knowledge Graph represents books, chapters, pages, topics and prerequisite relationships giving the system structured curriculum information for navigation and determining learning sequences. The Knowledge Graph finally includes 6 books, 71 chapters and 437 topics. The final data set consists of cleaned pages of the textbook, chunks of text, embedding indexes, and Knowledge Graph files used by the system backend.

IV. System Overview and Architecture

A. System Overview

SyllabusSherlock is a curriculum-aware learning platform designed to support students studying NCTB Mathematics and Science for Classes 6–8. The system combines a Knowledge Graph (KG), Retrieval-Augmented Generation (RAG), adaptive assessment and AI-assisted tutoring within a web-based learning environment. The Knowledge Graph organizes the curriculum into books, chapters, pages and topics while representing prerequisite relationships between concepts. This structure allows the system to provide a learning path rather than treating textbook content as an unstructured collection of documents.

When a student selects a class, subject, chapter and topic, the backend identifies the relevant curriculum structure and retrieves supporting textbook content through semantic retrieval. The retrieved content with relevant Knowledge Graph information is provided to the language model to generate curriculum-grounded responses to student questions. Student interactions with learning activities and assessments are also used to estimate topic mastery and support subsequent learning decisions. In this way, SyllabusSherlock connects curriculum structure, textbook evidence, AI-assisted learning and adaptive progress tracking within a single platform.

B. User and Core Features

SyllabusSherlock is primarily designed for students who interact with the system through a web based learning interface. Students can select their class and subject, navigate through chapters and topics, follow prerequisite-based learning paths, receive AI-assisted explanations, complete assessments and track their learning progress. The core features are:

- Curriculum Navigation: Students can navigate NCTB content from their class and subject to chapters and individual topics.
- Knowledge Graph Roadmap: Topics are presented as an interactive roadmap with connections representing relationships between topics and their prerequisites. The roadmap indicates a student's learning state through completed, current, available and locked topics.
- Prerequisite-based Learning: Prerequisite relationships determine topic availability allowing students to progress from foundational concepts to dependent topics and follow an appropriate learning sequence.
- AI Tutor: Students can ask questions about the topics they are learning and receive explanations based on relevant textbook information.
- Quiz and Assessment: Students can select topics and complete a limited number of quizzes to get their understanding. Practice questions are integrated into the learning process and can continue until sufficient confidence is reached with prerequisite relationships considered when selecting practice content.
- Progress and Gamification: Student progress is tracked through learning activities. While XP, streaks, rewards and companions encourage students to continue learning.

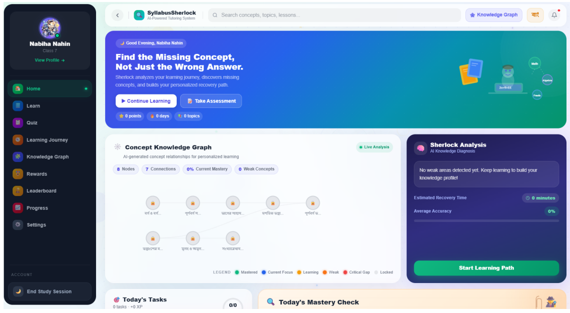

Fig. 1. Dashboard with knowledge graph, Sherlock AI, Learning Path, Today's tasks

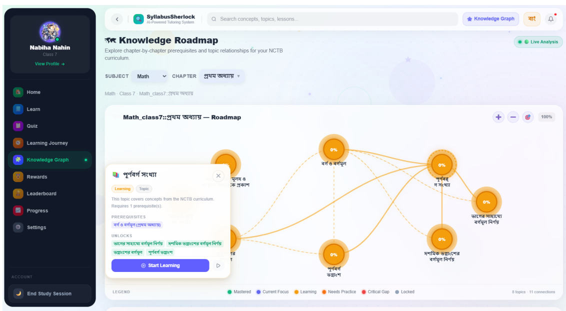

Fig. 2. Knowledge Graph–Based Curriculum Roadmap

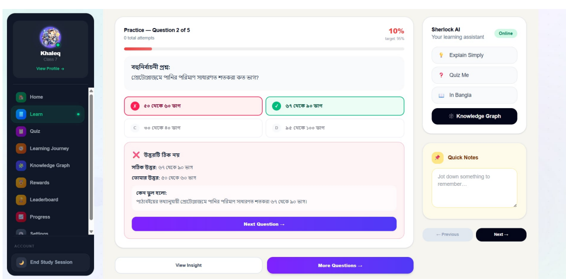

Fig. 3. Adaptive Quiz and Immediate Feedback

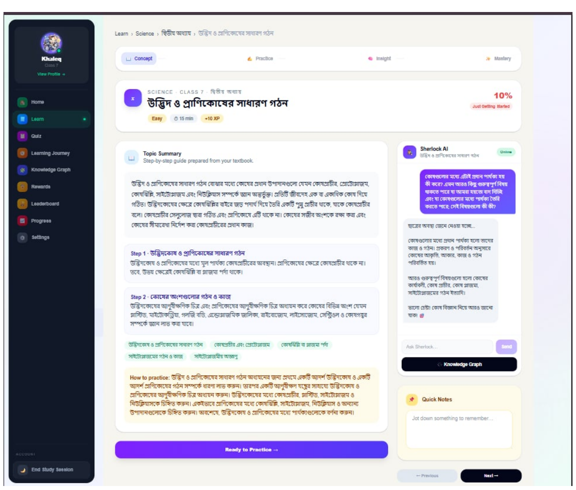

Fig. 4. Topic Learning Page with Sherlock AI Support

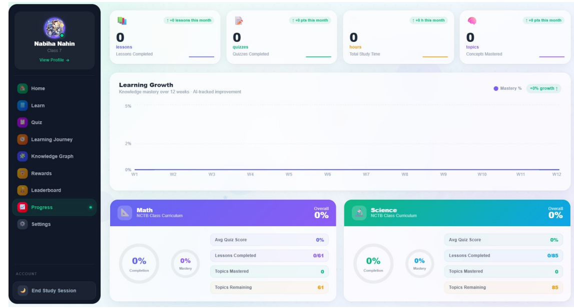

Fig. 5. Student Progress Page Showing Learning Growth, Assessment Performance, Study Habits, Sherlock's Learning insights

C. System Architecture

SyllabusSherlock follows a modular architecture that combines textbook processing, KG construction, semantic retrieval, backend services, AI generation, adaptive learning and a student friendly web interface as shown in Figure 2. NCTB textbooks are first processed using OCR and text preprocessing to obtain structured curriculum data and searchable textbook content. The Knowledge Graph represents relationships

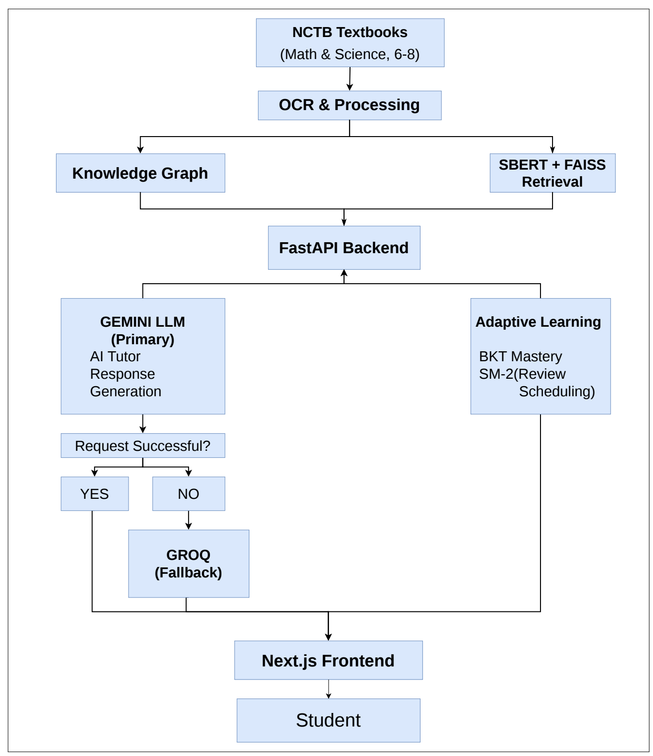

Fig. 6. Overall system architecture of SyllabusSherlock.

among books, chapters, pages and topics including prerequisite dependencies. Textbook context is embedded using SBERT and indexed with FAISS for semantic retrieval. The FastAPI backend coordinates these components including Gemini as the primary LLM, with Groq and xAI as fallback providers. BKT is used to estimate topic mastery from student assessment interactions while APScheduler supports review scheduling. The Next.js frontend provides access to the resulting learning services including curriculum navigation, the KG roadmap, AI-assisted learning, assessments and progress tracking.

D. Feature Workflows

SyllabusSherlock has three main workflows: curriculum navigation, AI assisted learning and adaptive assessment. When a student selects a topic, the Knowledge Graph provides the relevant textbook structure and prerequisite information to determine the next available topics.

For AI-assisted learning, the backend uses the selected topic and relevant curriculum context to retrieve textbook content through SBERT and FAISS. The retrieved content is provided to Gemini to generate a curriculum-grounded response and Grok is used as a fallback when the primary model is unavailable.

For assessment, students can select a topic and complete a limited quiz. After that, their performance is used to estimate topic mastery. During learning, practice questions provide additional opportunities for understanding and continue until sufficient confidence is reached while prerequisite relationships guide the practice sequence.

V. Evaluation: LLM as Judge

A. Evaluation Setup and Rubric

Evaluating SyllabusSherlock with an LLM-as-a-Judge approach focused on generating tutoring responses of high quality and aligned with the curriculum. NotebookLM and Claude were two independent judges for complementary evaluation. In all cases, each generated response was judged for its accuracy and matching with student's question, text book support, relevance, clarity, personalization and total marks against the evidence from the NCTB textbooks. A score of 1 to 5 was given to each criterion with 5 being the highest score. The accuracy in the fact and math content is assessed in terms of correctness and consistency of the retrieved curriculum content is assessed in terms of textbook grounding. Relevance and clarity evaluate the direct answer to the question asked and the clarity of the explanation. Personalization compares the response to the student's level of confusion, learning needs or type of explanation requested. Each judge additionally indicated pass/fail status and critical errors along with brief (qualitative) reasoning. Responses were classified as direct, explanation, or personalization to analyze performance by tutoring behaviors. Lastly, score, pass/fail and critical-error agreement were used to compare the two independent judgments. The comparison was not made to either judge as ground truth but to be used to give additional evidence of the quality of the responses and the consistency of the evaluations.

B. Overall Evaluation Results

Both LLM judges rated the generated responses highly in terms of correctness and textbook grounding. NotebookLM achieved a mean overall score of 4.78/5 with 92.68 % of responses passing, while Claude obtained 4.27/5 with an 85.37 % pass rate. The highest scores were observed for correctness and textbook grounding, whereas personalization received comparatively lower scores indicating room for improvement in adapting explanations to individual learning needs. The two judges show a consistent positive assessment of the system, while their lower personalization scores suggest that future improvements should focus on producing responses that more explicitly address students' stated difficulties and learning context.

TABLE I
Comparison of LLM-Judge Evaluation Results

Criterion	NotebookLM	Claude
Correctness	4.854	4.683
Textbook grounding	4.732	4.610
Relevance	4.976	4.439
Clarity	4.902	4.463
Personalization	3.561	2.805
Overall score	**4.781**	**4.268**
Pass rate	92.68%	85.37%
Critical-error rate	2.44%	4.88%

C. Inter-Judge Agreement

NotebookLM and Claude showed partial but meaningful agreement in evaluating the generated responses. The overall-score agreement was 65.85% exact increasing to 80.49% when a difference of ± 1 point was allowed. Their pass/fail decisions agreed on 87.80% of cases with a Cohen's kappa of 0.384 while critical-error judgments agreed on 97.56% of cases ($\kappa = 0.656$). Claude consistently assigned lower scores across all six criteria, with the largest difference occurring in personalization (3.56 vs. 2.80). This suggests that personalization was the most subjective dimension, while correctness and textbook grounding showed stronger consistency between the two judges. These results indicate that the judges provide complementary assessments rather than establishing a single ground-truth evaluation.

TABLE II
Agreement Between LLM Judges

Measure	Value
Overall-score exact agreement	65.85%
Overall-score agreement (± 1)	80.49%
Pass/fail agreement	87.80%
Pass/fail Cohen's κ	0.384
Critical-error agreement	97.56%
Critical-error Cohen's κ	0.656

D. Error Analysis and Discussion

The evaluation identified a small number of notable weaknesses. The main issues occurred in personalization, where responses sometimes failed to address the learner's specific confusion or omitted requested step-by-step explanations. Two critical errors were also identified, involving incorrect comma placement in the international numbering system and incorrect rounding of a square-root approximation. These cases show that, although the system generally produces correct and textbook-grounded responses, personalized explanations remain more error-prone than direct factual responses. The lower personalization scores from both judges further support the need for improved handling of learner-specific misconceptions and multi-step explanations.

VI. Discussion

One application benefit of the system is its modularity. The FastAPI backend isolates the curriculum information, knowledge retrieval, interaction with the LLM and adaptive learning. Primary language model is Gemini and fallback will be Groq. This backup enables the AI learning feature to function when the main model is unavailable. The quality of the generated responses still depends on the quality of the retrieved textbook context and the language model.

The existing system also has some drawbacks. Currently, six NCTB textbooks are included in the dataset for the subjects Mathematics and Science for Classes 6-8 and a limited number of manually structured prerequisite relationships are included in the Knowledge Graph. Furthermore, the adaptive learning piece relies on data from student interactions, and increased student interaction data can be beneficial in generating accurate estimates of student mastery for a longer duration of use.

In summary, SyllabusSherlock combines structured curriculum relationships, retrieval based on textbook content, AI generative capabilities and adaptive learning into a single platform to enable curriculum-aware tutoring.

VII. Conclusion and Future Work

SyllabusSherlock is a working prototype of curriculum-aware learning based on NCTB textbooks. The completed system supports structured curriculum navigation, prerequisite-based learning, textbook-grounded AI assistance, quizzes, practice activities and adaptive progress tracking within a single platform. The present implementation is for the subject areas of Mathematics and Science in Classes 6-8 and it is designed to be developed and evaluated further.

Future development will be on extending Mathematics and Science coverage to more subjects and classes. The Knowledge Graph can further be expanded using additional correctly validated relationships that are deemed as prerequisite knowledge for the learning-path recommendations. The adaptive learning part can be further optimized through more student interaction data and more accurate estimation of mastery and scheduling of student review over longer use periods. Further, larger questions sets, student studies conducted by humans and human validation of AI generated answers can be incorporated into future assessments to explore the effectiveness of learning, usability and engagement. Other enhancements may explore more powerful retrieval methods and LLM variants to improve the uniformity and curriculum of AI-supported responses.

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